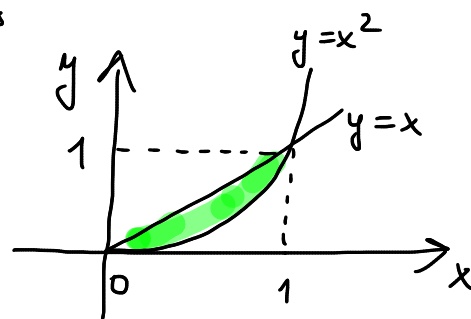


Bonus 4 - řešení!

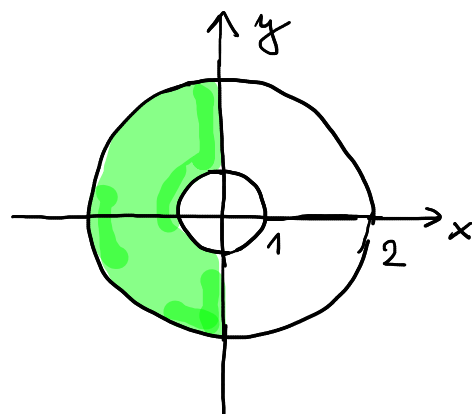
① $\iint_{\Omega} (x^2 y + 1) dx dy, \Omega:$



a) $\int_0^1 \left[\int_{x^2}^x (x^2 y + 1) dy \right] dx$
 $\int_0^1 \left[\int_y^{\sqrt{y}} (x^2 y + 1) dx \right] dy$

b) $\int_0^1 \left[\int_{x^2}^x (x^2 y + 1) dy \right] dx = \int_0^1 \left[\frac{x^2 y^2}{2} + y \right]_{x^2}^x dx$
 $= \int_0^1 \left(\frac{x^4}{2} + x - \left(\frac{x^6}{2} + x^2 \right) \right) dx$
 $= \left[\frac{x^5}{10} + \frac{x^2}{2} - \frac{x^7}{14} - \frac{x^3}{3} \right]_0^1 = \frac{1}{10} + \frac{1}{2} - \frac{1}{14} - \frac{1}{3} = \underline{\underline{\frac{41}{210}}}$

② $\iint_{\Omega} \frac{2y + \sqrt{x^2 + y^2}}{x} dx dy, \Omega:$



$\frac{\pi}{2} \leq \varphi \leq \frac{3\pi}{2}$
 $1 \leq r \leq 2$

$\Rightarrow \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \left[\int_1^2 \frac{2n \cdot \sin \varphi + \sqrt{n^2}}{n \cdot \cos \varphi} \cdot n dr \right] d\varphi$

$= \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \left[\int_1^2 \frac{2n \cdot \sin \varphi + n}{\cos \varphi} dr \right] d\varphi$

$= \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{2 \sin \varphi + 1}{\cos \varphi} d\varphi \cdot \int_1^2 r dr$